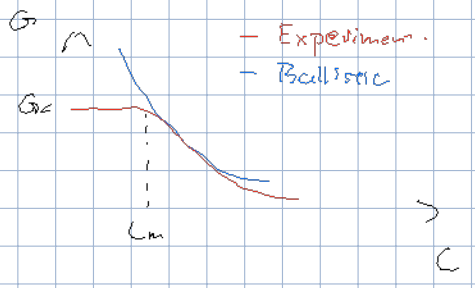
§ 2 Landauer - Buttiker Method.

§ 2.1 Resistance of a ballistic conductor



1. Ballistic / Diffusive

if No scattering,

scaling : $G = \sigma \frac{W}{L}$

2. Contact Resistance. (G_c^{-1})

$L \to 0$. why not $G \to \infty$?

1. Buttiker Formula

$$I_p = \sum_{q \neq p} G_{pq} (V_p - V_q)$$

$$V_p = \frac{\sum_{q \neq p} G_{pq} V_q}{\sum_{q \neq p} G_{pq}}$$

2. Reciprocity :

Onsager Relation : $\rho_{ij}(B) = \rho_{ji}(-B)$

$$\rho(B) = \begin{pmatrix} \rho_{xx} & \rho_{xy} \\ \rho_{yx} & \rho_{yy} \end{pmatrix} = \begin{pmatrix} \rho_{xx} & \frac{1}{2}(\rho_{xy} + \rho_{yx}) \\ \frac{1}{2}(\rho_{yx} + \rho_{xy}) & \rho_{yy} \end{pmatrix} + \begin{pmatrix} 0 & \frac{1}{2}(\rho_{xy} - \rho_{yx}) \\ \frac{1}{2}(\rho_{yx} - \rho_{xy}) & 0 \end{pmatrix}$$

$$= \begin{pmatrix} \rho_{xx}(B) & \frac{1}{2}\left[\rho_{xy}(B) + \rho_{xy}(-B)\right] \\ \frac{1}{2}\left[\rho_{yx}(B) + \rho_{yx}(-B)\right] & \rho_{yy}(B) \end{pmatrix} + \begin{pmatrix} 0 & \frac{1}{2}\left[\rho_{xy}(B) - \rho_{xy}(-B)\right] \\ \frac{1}{2}\left[\rho_{yx}(B) - \rho_{yx}(-B)\right] & 0 \end{pmatrix}$$

$$= \rho_S(B) + \rho_A(B)$$

$$\begin{cases} \rho_S(B) = \rho_S(-B) \\ \rho_A(B) = -\rho_A(-B) \end{cases} \qquad \rho_{xx}(B) = \rho_{xx}(-B) \rightarrow R_{ab}(B) = R_{ba}(-B)$$

$$[I] = [G][V] \qquad [G] = [R^{-1}] = \begin{pmatrix} \ddots & & -G_{ap} \\ & \sum_{q \neq p} G_{pq} & \\ -G_{ap} & & \ddots \end{pmatrix}$$

① off-grid

$G_{ap}(B) = G_{pa}(-B)$

$\hookrightarrow R^{-1}_{ap}(B) = R^{-1}_{pa}(-B)$

② diag

$\sum_q G_{pq} = \sum_q G_{qp} \rightarrow \sum_{q \neq p} G_{pq}(B) = \sum_{q \neq p} G_{qp}(B)$

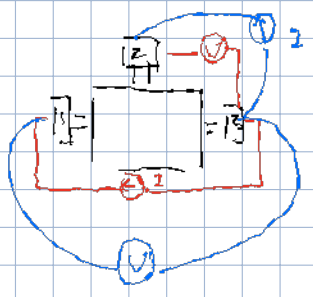
$G_{pp}(B) = \sum_{q \neq p} G_{qp}(B) = \sum_{q \neq p} G_{pq}(B) = \sum_{q \neq p} G_{pq}(-B) = G_{pp}(-B)$

$\hookrightarrow R^{-1}_{pp}(B) = R^{-1}_{pp}(-B)$

$\Rightarrow R^{-1}(B) = (R^{-1})^T(-B) \rightarrow R(B) = R^T(-B)$

① 3-Terminal

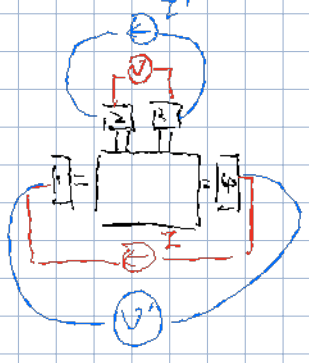
$R_{34}(B) = R_{21}(B) = R_{12}(-B) = R_{32}(-B)$



② 4-Terminal

$R_{14}(B) = R_{21}(B) - R_{31}(B)$

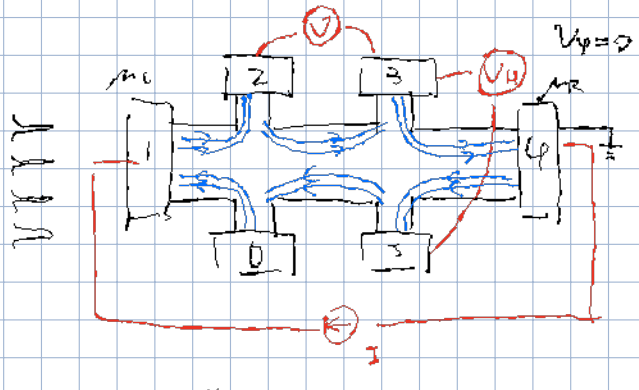
$= R_{12}(-B) - R_{13}(-B) = R_{14}'(-B)$



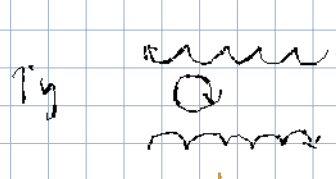
3. Application of LB

(1) QHE





$\nu = 2 \Rightarrow$ 2 edge state



$\vec{k} = k_x \hat{x}$



Chiral

G_{pq} ($p \leftarrow q$)?

$G_{xx} = \frac{2e^2}{h} M = \frac{\nu e^2}{h}$

$$= \begin{pmatrix} G_{xx} & & & & G_{xx} \\ G_{xx} & G_{xx} & & & \\ & G_{xx} & G_{xx} & & \\ & & G_{xx} & G_{xx} & \\ & & & G_{xx} & G_{xx} \end{pmatrix}$$

$V_4 = 0$

$$I_4 = \begin{pmatrix} I_1 \\ \cdot \\ \cdot \\ \cdot \\ I_5 \end{pmatrix} = \begin{pmatrix} G_{xx} & & & & -G_{xx} \\ -G_{xx} & G_{xx} & & & \\ & -G_{xx} & G_{xx} & & \\ & & -G_{xx} & G_{xx} & \\ & & & -G_{xx} & G_{xx} \end{pmatrix} \begin{pmatrix} V_1 \\ \cdot \\ \cdot \\ \cdot \\ V_5 \end{pmatrix}$$

$I_2 = I_3 = I_5 = I_6 = 0$

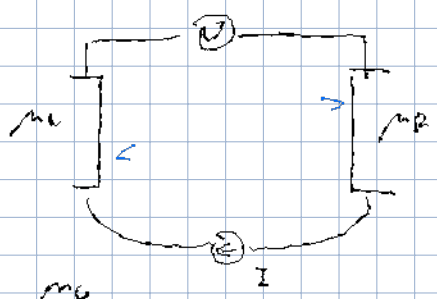
$V_1 = V_2 = V_3, \quad V_5 = V_6 = 0 = V_4$

$I_1 = G_{xx} V_1, \quad I_4 = -I_1$

$R_L = \frac{V_2 - V_3}{I_1} = \frac{V_6 - V_5}{I_1} = 0$

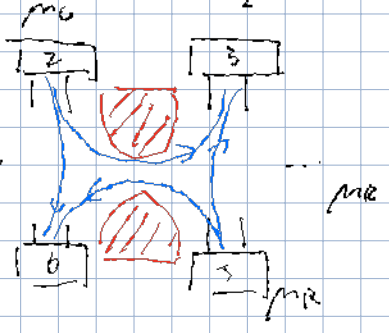
$R_H = \frac{V_2 - V_6}{I_1} = \frac{V_1 - V_5}{I_1} = \frac{1}{G_{xx}} = \frac{h}{\nu e^2}$

$-G_c \quad G_c$ V_6 Quantized:



$$R = \frac{V}{I} = \frac{h}{e^2 n}$$





$N < n$ 个通道完全透射，$n-N$ 个反射。

$$I_1 = \frac{2e^2}{h} \frac{V_1}{n} = \frac{2e^2}{h} n \cdot \frac{N}{n} V_1 = \frac{2e^2}{h} N V_1$$

反射率 $p = 1 - \frac{N}{n}$

$$V_6 = \frac{(n-N) G_{6L} V_2 + N G_{6R} V_5}{(n-N) G_{6L} + N G_{6R}}$$

$G_{pq} = G_c$

$V_2 = V_1, \quad V_4 = V_6 = 0$

$$= \frac{n-N}{n} V_1$$

$$V_5 = \frac{N V_2 + (n-N) V_5}{n} = \frac{N}{n} V_1$$

$$R_H = \frac{V_2 - V_5}{I_1} = \frac{n-N}{n} \frac{V_1}{I_1} = \frac{h}{2e^2} \left(\frac{1}{N} - \frac{1}{n} \right)$$

$$R_H = \frac{V_2 - V_6}{I_1} = \frac{N}{n} \frac{V_1}{I_1} = \frac{h}{2e^2} \frac{1}{n}$$

G_{pq}

$$= \begin{pmatrix} 0 & & & & G_c \\ G_c & & & & 0 \\ & p G_c & & (1-p) G_c & \\ & & G_c & & \\ & & & G_c & \end{pmatrix}$$

$$= \begin{pmatrix} 0 & (1-p) G_c & & p G_c & 0 \\ & & & & \end{pmatrix}$$

$V_1 = V_2, \quad V_3 = V_4 = 0$

$I_2 = I_3 = I_5 = I_6$

$$\begin{pmatrix} I_1 \\ I_2 \\ I_3 \\ I_5 \\ I_6 \end{pmatrix} = \begin{pmatrix} G_c & & & -G_c & \\ -G_c & G_c & & & \\ -(1-p) G_c & G_c & -p G_c & & \\ & & G_c & & \\ & & & & \end{pmatrix} \begin{pmatrix} V_1 \\ V_1 \\ V_3 \\ V_5 \\ V_6 \end{pmatrix}$$

$$\begin{cases} -(1-p) G_c V_1 + G_c V_3 = 0 \\ -p G_c V_3 + G_c V_6 = 0 \end{cases}$$

$\rightarrow V_6 = p V_3 = p V_1$

$I_1 = G_c V_1 - G_c V_6 = (1-p) G_c V_1$

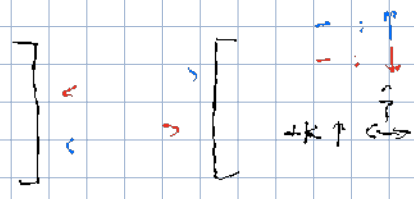
$R = \frac{V_2 - V_3}{I_1} = \frac{V_6 - V_5}{I_1} = \frac{p V_1}{(1-p) G_c V_1} = \frac{p}{(1-p) G_c}$

$R_H = \frac{V_2 - V_0}{I_1} = (1-p)\frac{V_1}{I_1} = \frac{1}{G_{xy}}$

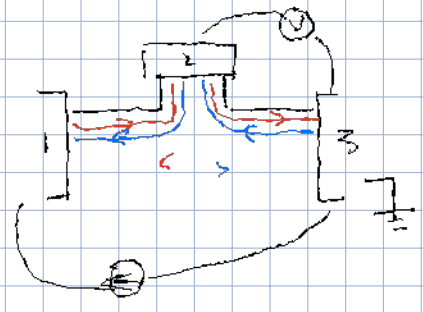
E 2.3. E 2.4. E 4.1 E 4.2

(2) QSH

$\hat{T}^{-1} H_{QSH} \hat{T} = H_{QSH}$



$\vec{k}$ 축 대칭.

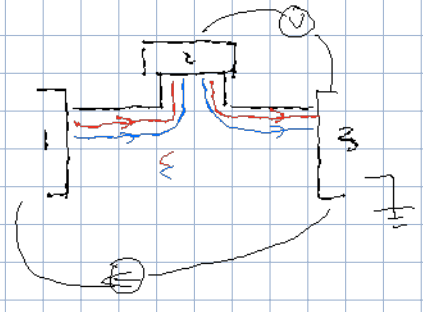


$G_{spin} = \begin{pmatrix} 0 & G_{xy} & G_{xz} \\ G_{yx} & 0 & G_{yz} \\ G_{zx} & G_{zy} & 0 \end{pmatrix}$ $G_{xy} = \frac{e^2}{h}$

$\begin{pmatrix} I_1 \\ I_2 \end{pmatrix} = \begin{pmatrix} 2G_{xy} & -G_{xy} \\ -G_{xy} & 2G_{xy} \end{pmatrix} \begin{pmatrix} V_1 \\ V_2 \end{pmatrix}$

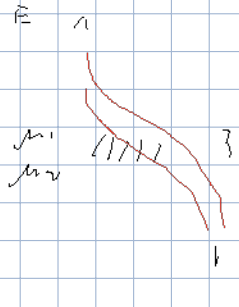
$I_2 = 0$: $\begin{cases} 2G_{xy}V_1 - G_{xy}V_2 = I_1 \\ -G_{xy}V_1 + 2G_{xy}V_2 = 0 \end{cases} \rightarrow \begin{cases} I_1 = \frac{3}{2}G_{xy}V_1 \\ V_2 = \frac{1}{2}V_1 \end{cases}$

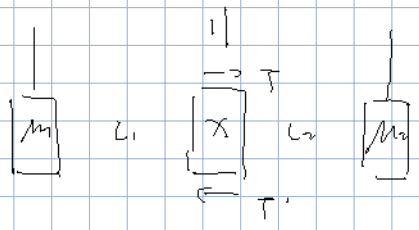
$\Rightarrow G_H = \frac{I_1}{V_1} = \frac{3}{2}G_{xy}$



§ 2.5 Non-zero T & bias

1. $T = 0$ $T \neq 0$





$i_1^+ \rightarrow$ $i_2^+ \rightarrow$

$i_1^- \leftarrow$ $i_2^- \leftarrow$

Influx from 1 : $i_1^+(E) = \frac{q}{2\pi} v \cdot f_1(E)$

2 : $i_2^-(E) = \frac{q}{2\pi} v \cdot f_2(E)$

Outflux from 1 : $i_1^-(E) = (1-T) i_1^+(E) + T i_2^-(E)$

2 : $i_2^+(E) = T i_1^+(E) + (1-T) i_2^-(E)$

Net current $i(E) = i_1^+(E) - i_1^-(E) = i_2^-(E) - i_2^+(E)$

$= T i_1^+(E) - T i_2^-(E)$

$= \frac{q}{2\pi} v(E) \left[T(E) f_1(E) - T(E) f_2(E) \right]$

$= \frac{q}{2\pi} \left[\bar{T}(E) f_1(E) - \bar{T}(E) f_2(E) \right]$ $\quad \bar{T} = M T$

Total current $I(E) = \int i(E) dE$

$= \frac{q}{2\pi} \int \bar{T}(E) \left[f_1(E) - f_2(E) \right] dE$ $\quad (\bar{T} = \bar{T}')$

2. Linear response. $I = G|V| \propto V$

bias : $\Delta\mu = \mu_1 - \mu_2$

$\Delta I = \int \frac{q}{2\pi} \bar{T}(E) \left[f_1(E) - f_2(E) \right] dE = I_S$

$\Delta(f_1 - f_2) \approx \frac{\partial f}{\partial E} \Delta\mu = -\frac{\partial f_0}{\partial E} \Delta\mu$ $\quad f_1(E) = f(E, \mu_1) = \frac{1}{1 + e^{(E-\mu_1)/kT}}$

$G = \frac{I_S}{\Delta\mu} = \frac{q^2}{2\pi} \int \bar{T}(E) \left(-\frac{\partial f_0}{\partial E} \right) dE$

$T = 0$ K : $f_0(E) = \theta(\mu - E)$, $-\frac{\partial f_0}{\partial E} = \delta(E - \mu)$

$G_1 = \frac{2e^2}{h}\,\bar{T}(\mu)$

Linear condition : $\bar{T}(E) \approx$ Const. $\quad E \in \mu_1, \mu_2 \pm kT$

$$I = \frac{2e}{h}\,\bar{T}(E_F)\int (f_1 - f_2)\,dE \simeq \frac{2e}{h}\,\bar{T}(E_F)(\mu_1 - \mu_2)$$

Generally,

$$f_1(E) - f_2(E) = \int_{\mu_2}^{\mu_1} \frac{\partial}{\partial E'}\left(\frac{1}{e^{(E-E')/kT}+1}\right) dE'$$

$$= -\int_{\mu_2}^{\mu_1} \frac{\partial}{\partial(E-E')}\left(\frac{1}{e^{(E-E')/kT}+1}\right) dE'$$

$\bar{F}_T(E-E_F)$

$$\tilde{f}_0(E) = \frac{1}{e^{E/kT}+1}$$

$$\bar{F}_T(E) = -\frac{\partial}{\partial E}\tilde{f}_0(E) = \frac{1}{4kT}\operatorname{sech}^2\left(\frac{E}{2kT}\right)$$

kT

$$= \int_{\mu_2}^{\mu_1} \bar{F}_T(E-E')\,dE'$$

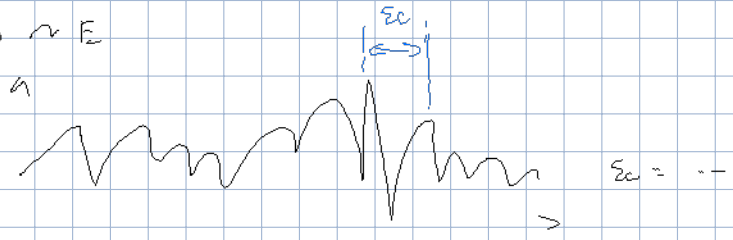
$$I = \frac{2e}{h}\int \bar{T}(E)\left(\int_{\mu_2}^{\mu_1} \bar{F}_T(E-E')\,dE'\right) dE$$

$$= \frac{1}{e}\int_{\mu_2}^{\mu_1}\left[\frac{2e^2}{h}\int \bar{T}(E)\,\bar{F}_T(E-E')\,dE\right] dE'$$

$$= \frac{1}{e}\int_{\mu_2}^{\mu_1} G(E')\,dE'$$

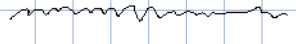
$\Rightarrow G(E) \approx$ Const. $= G(E_F)$, $E \in (\mu_2, \mu_1)$

$\bar{T}(E) \sim E$



$\varepsilon_c =$ --

(1) sharpe $\bar{T}(E)$, small ε_c



if $\varepsilon_c \ll kT$ -> $\bar{T}(E)\,\bar{F}_T(E-E')$ 平滑

③ smooth $T(E)$, large Σ_c

$\Delta\mu \ll \Sigma_c$

$\Rightarrow$ $\Delta\mu \ll k_B T$?

3. multi-terminal

$$i_p(E) = \frac{q}{h} \sum_q \left\{ \bar{T}_{qp}(E) f_p(E) - \bar{T}_{pq}(E) f_q(E) \right\}$$

$$= \frac{q}{h} \sum_{q \neq p} \left\{ \bar{T}_{qp}(E) f_p(E) - \bar{T}_{pq}(E) f_q(E) \right\}$$

平衡) $\forall q$ $\rightarrow$ $i_p = 0$ $\Rightarrow$ $\sum_q \bar{T}_{qp}(E) = \sum_q \bar{T}_{pq}(E)$

?

$$i_p(E) = \frac{q}{h} \sum_{q \neq p} \bar{T}_{pq}(E) \left\{ f_p(E) - f_q(E) \right\}$$

$$I_p = \int i_p(E)\, dE$$

$$= \frac{q}{h} \int \bar{T}_{pq}(E) \left\{ f_p(E) - f_q(E) \right\} dE$$

$$= \frac{q}{h} \int \bar{T}_{pq}(E) \left\{ \int_{\mu_q}^{\mu_p} F_T(E - E')\, dE' \right\} dE$$

$$= \frac{1}{e} \int_{\mu_q}^{\mu_p} \left\{ \frac{2e^2}{h} \int \bar{T}_{pq}(E)\, F_T(E - E')\, dE \right\} dE'$$

$$= \frac{1}{e} \int_{\mu_q}^{\mu_p} G_{pq}(E')\, dE$$

$\Rightarrow$ $G_{pq}(E') \cong$ const. $= G(E_F)$, $E \in [\mu_q, \mu_p]$

$$I_p = \sum_q I_{pq} = \frac{1}{e} \sum_q G_{pq}(E_F)(\mu_p - \mu_q) = G_{pq}(E_F)(V_p - V_q)$$

§ 2.6 Exclusion Principle.

$$i_p(E) = \frac{2e}{h} \sum_q \left(\bar{T}_{qp}(E) f_p(E) - \bar{T}_{pq}(E) f_q(E) \right)$$

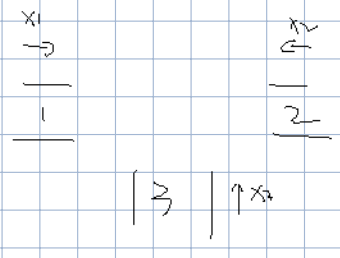
X 不能用微扰理论

$$i_p'(E) = \frac{2e}{h} \sum_q \bar{T}_{qp}(E) f_p(E) \{1 - f_q(E)\} - \bar{T}_{pq}(E) f_q(E) \{1 - f_p(E)\}$$

$$i_p'(E) - i_p(E) = \frac{2e}{h} \sum_q \left(\bar{T}_{pq}(E) - \bar{T}_{qp}(E) \right) f_q(E) f_p(E)$$

① Scattering States.

p 中 q 散射态 (q, k)



$$\psi_p(q) = S_{pq} \chi_p^+(y_p) e^{ik_p^+ x_p} + S_{pq}' \chi_p'(y_p) e^{ik_p^- x_p}$$

$$\downarrow$$

$$i_p(q) = \frac{2e}{h} (S_{pq} - T_{pq})$$

$$I_p = \int \sum_q f_q(E) i_p(q) dE$$

$$= \frac{2e}{h} \int (f_p - \sum_q f_q T_{pq}) dE \qquad \sum_q T_{pq} = \sum_q T_{qp} = 1$$

$$= \frac{2e}{h} \int \sum_q T_{pq} (f_p - f_q) dE$$